MINI PROJECT 2

Name:Prathiksha Harish Baleri

USN:4CB22CB044

Domain: Data science

Iris dataset

The Iris dataset is contains samples of iris flowers across three variety—Setosa, Versicolor, and Virginica—with four measured features: sepal length, sepal width, petal length, and petal width

Use of Linear regression

We upload the dataset in jupyter notebook

```
[1]: import pandas as pd
```

```
[2]: iris=pd.read_csv('iris.csv')
```

```
[4]: iris.head()
```

[4]:

	sepal.length	sepal.width	petal.length	petal.width	variety
0	5.1	3.5	1.4	0.2	Setosa
1	4.9	3.0	1.4	0.2	Setosa
2	4.7	3.2	1.3	0.2	Setosa
3	4.6	3.1	1.5	0.2	Setosa
4	5.0	3.6	1.4	0.2	Setosa

This shows the top 5 data in the dataset this basically shows that the dataset is uploaded and read

```
[8]: iris[iris['sepal.width']>4]
```

[8]:

	sepal.length	sepal.width	petal.length	petal.width	variety
15	5.7	4.4	1.5	0.4	Setosa
32	5.2	4.1	1.5	0.1	Setosa
33	5.5	4.2	1.4	0.2	Setosa

This shows the data where sepal width >4 the variety is setosa

```
[12]: iris[iris['petal.width']>1]
```

[12]:

	sepal.length	sepal.width	petal.length	petal.width	variety
50	7.0	3.2	4.7	1.4	Versicolor
51	6.4	3.2	4.5	1.5	Versicolor
52	6.9	3.1	4.9	1.5	Versicolor
53	5.5	2.3	4.0	1.3	Versicolor
54	6.5	2.8	4.6	1.5	Versicolor
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	Virginica
146	6.3	2.5	5.0	1.9	Virginica
147	6.5	3.0	5.2	2.0	Virginica
148	6.2	3.4	5.4	2.3	Virginica
149	5.9	3.0	5.1	1.8	Virginica

93 rows × 5 columns

Here data with petal width >1 is shown which implies tht versicolor and virginica have petal width>1

```
[13]: iris[iris['petal.width']>2]
```

[13]:

	sepal.length	sepal.width	petal.length	petal.width	variety
100	6.3	3.3	6.0	2.5	Virginica
102	7.1	3.0	5.9	2.1	Virginica
104	6.5	3.0	5.8	2.2	Virginica
105	7.6	3.0	6.6	2.1	Virginica
109	7.2	3.6	6.1	2.5	Virginica
112	6.8	3.0	5.5	2.1	Virginica
114	5.8	2.8	5.1	2.4	Virginica
115	6.4	3.2	5.3	2.3	Virginica
117	7.7	3.8	6.7	2.2	Virginica
118	7.7	2.6	6.9	2.3	Virginica
120	6.9	3.2	5.7	2.3	Virginica
124	6.7	3.3	5.7	2.1	Virginica
128	6.4	2.8	5.6	2.1	Virginica
132	6.4	2.8	5.6	2.2	Virginica
135	7.7	3.0	6.1	2.3	Virginica
136	6.3	3.4	5.6	2.4	Virginica
139	6.9	3.1	5.4	2.1	Virginica
140	6.7	3.1	5.6	2.4	Virginica
141	6.9	3.1	5.1	2.3	Virginica
143	6.8	3.2	5.9	2.3	Virginica
144	6.7	3.3	5.7	2.5	Virginica
145	6.7	3.0	5.2	2.3	Virginica
148	6.2	3.4	5.4	2.3	Virginica

The variety with petal width>2 is virginica

```
[46]: y=iris[['sepal.length']]
```

```
[47]: x=iris[['sepal.width','petal.length']]
```

```
[48]: from sklearn.model_selection import train_test_split;
```

```
[49]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
```

```
[50]: x_train.head()
```

[50]:

	sepal.width	petal.length
9	3.1	1.5
133	2.8	5.1
141	3.1	5.1
140	3.1	5.6
45	3.0	1.4

```
[51]: x_test.head()
```

[51]:

	sepal.width	petal.length
66	3.0	4.5
138	3.0	4.8
27	3.5	1.5
93	2.3	3.3
84	3.0	4.5

```
[52]: y_train.head()
```

[52]:

	sepal.length
9	4.9
133	6.3
141	6.9
140	6.7
45	4.8

```
[53]: y_test.head()
```


```
[52]: y_train.head()
```

[52]:

	sepal.length
9	4.9
133	6.3
141	6.9
140	6.7
45	4.8

```
[53]: y_test.head()
```


[53]:

	sepal.length
66	5.6
138	6.0
27	5.2
93	5.0
84	5.4

```
[54]: #categorize the data was done till here as train and test data
```

```
[55]: from sklearn.linear_model import LinearRegression
```

```
[56]: lr=LinearRegression()
```

```
[57]: lr.fit(x_train,y_train)
```

```
[57]: ▾ LinearRegression
      LinearRegression()
```

```
[58]: y_pred=lr.predict(x_test)
```

```
[59]: y_test.head()
      #actual value
```

[59]:

	sepal.length
66	5.6
138	6.0
27	5.2

```
[56]: lr=LinearRegression()

[57]: lr.fit(x_train,y_train)

[57]: LinearRegression
      LinearRegression()

[58]: y_pred=lr.predict(x_test)

[59]: y_test.head()
      #actual value
```

[59]:

	sepal.length
66	5.6
138	6.0
27	5.2
93	5.0
84	5.4

```
[60]: y_pred[0:5]
      #predicted value

[60]: array([[6.17251753],
             [6.31334815],
             [5.04374297],
             [5.21785076],
             [6.17251753]])

[61]: #we find out the error in prediction
      from sklearn.metrics import mean_squared_error
      mean_squared_error(y_test,y_pred)

[61]: 0.11754920595342445
```

This is of 2 independent variables sepal length and petal width

Model 2 is for 2 independent variables

Here I used petal width and sepal width

model 2

```
[62]: #using 3 independent values
[63]: y=iris[['sepal.length']]
[64]: x=iris[['sepal.width','petal.width']]
[65]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
[66]: lr2=LinearRegression()
[67]: lr2.fit(x_train,y_train)
[67]: LinearRegression
      LinearRegression()
[68]: y_pred=lr2.predict(x_test)
[69]: y_test.head()
```

[69]:

	sepal.length
115	6.4
85	6.0
15	5.7
32	5.2
5	5.4

```
[70]: y_pred[0:5]
[70]: array([[7.04408343],
             [6.41551789],
             [5.61756435],
             [5.18407347],
             [5.40483481]])
[71]: mean_squared_error(y_test,y_pred)
[71]: 0.17942057744469267
```

Model 3

I used 3 independent variable petal length and petal width

model 3

```
[72]: y=iris[['sepal.length']]
[73]: x=iris[['petal.length','petal.width']]
[74]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)
[75]: lr3=LinearRegression()
[76]: lr3.fit(x_train,y_train)
[76]: LinearRegression
      LinearRegression()
[77]: y_pred=lr3.predict(x_test)
```

```
[78]: y_test.head()
```

[78]:

	sepal.length
37	4.9
81	5.5
32	5.2
88	5.6
20	5.4

```
[79]: y_pred[0:5]

[79]: array([[4.85383615],
             [5.83937991],
             [4.90770345],
             [5.97038104],
             [4.98728203]])

[80]: mean_squared_error(y_test,y_pred)

[80]: 0.21969215145134882
```

Model 4

I used 3 independent variables and changed y to petal length

model 4

```
[81]: y=iris[['petal.length']]

[82]: x=iris[['sepal.length','petal.length','petal.width']]

[83]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)

[84]: lr4=LinearRegression()

[87]: lr4.fit(x_train,y_train)

[87]: LinearRegression
      LinearRegression()

[88]: y_pred=lr4.predict(x_test)

[89]: y_test.head()
      y_pred[0:5]

[89]: array([[1.7],
             [5.8],
             [1.3],
             [4.7],
             [1.7]])

[90]: mean_squared_error(y_test,y_pred)

[90]: 1.169048035931694e-30
```

Model 5 and model 6 I changed y=sepal.width and used 3 independent variables

model 5

```
[122]: y = iris[['sepal.width']]

[123]: x = iris[['sepal.length','petal.length','petal.width']]

[124]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)

[125]: lr5=LinearRegression()
       lr5.fit(x_train,y_train)
       y_pred=lr5.predict(x_test)
       y_test.head()
       mean_squared_error(y_test,y_pred)

[125]: 0.10002515686394756
```

model 6

```
[127]: y = iris[['sepal.width']]

[128]: x = iris[['petal.length','petal.width']]

[129]: x_train,x_test,y_train,y_test=train_test_split(x,y,test_size=0.3)

[130]: lr6=LinearRegression()
       lr6.fit(x_train,y_train)

[130]: LinearRegression
       LinearRegression()

[131]: y_pred=lr6.predict(x_test)
       y_test.head()
```

[131]:

	sepal.width
9	3.1
44	3.8
124	3.3
65	3.1
74	2.9

```
[132]: mean_squared_error(y_test,y_pred)

[132]: 0.14743157690949393
```

According to me Model 5 with mean squared error 0.10002515686394756

Will be having the best prediction with y=sepal width and 3 independent variables sepal length,petal length,petal width